

	Pre-thesis (Planning–Design)	Phase I Weeks 1–4	Phase II Weeks 5–8	Phase III Weeks 9–12	Phase IV Weeks 13–16
SDLC Stage Mapping	Planning – Design	Development	Development	Verification – Validation	Maintenance
Primary Focus	Requirements, Archit System & Smart Con Design, Dataset & ML	Core Contracts & Infrastructure Development	Lending Lifecycle, Oracle & ML Integration, Front-end & API	Testing, Security, Validation, Simulation, Documentation	Deployment Hardening, Monitoring, Optimization, Handover
Primary Deliverables	• SRS & Use Cases • 4-tier Architecture Design • Smart Contract Specifications • DB Schema & API Interface • ML Pipeline Design • Project Plan & Risk	• Four-tier Core Contracts (WB, NB, LB, Wallet Auth) • Role & Access Control • Wallet Integration • Basic Frontend Shell • DB Schema & Migration • Unit Tests (Core)	• Lending & Deposit Modules (Savings, Vault, Fixed Deposit, Group Lending Pool, IBL Upward Deposit) • Oracle Integration • ML Scoring API (RF + Gradient Boosting) • Authority Brief (SAP) • Frontend Flows (End-to-End)	• Integration & System Testing • Security Testing (Slither, Mythril) • Simulation (300 clients) • Formal Methods (Unverified) • Documentation (Technical) • Performance & Gas Analysis	• Testnet Deployment • Monitoring & Alerting • Bug Fixes & Optimization • Backup & Recovery • Final Report & Handover • Future Work Roadmap
Optional Agent (dashed)	—	—	—	Conversational Agent (MCP Prototype Development)	Agent Integration & Evaluation
Key Milestone Outcome	Design Complete (Spec Baseline)	Core Platform Operational	End-to-End Loan Flow with ML Scoring	Validated & Tested System (MVT Ready)	Deployed, Monitored & Handover Complete

